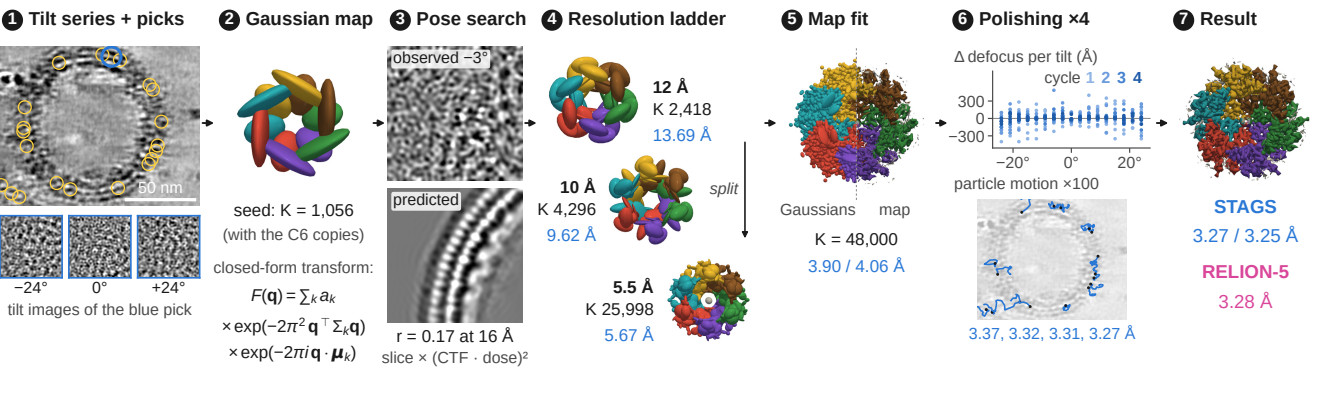

1 Tilt series + picks
60 nm
−24°
0°
+24°
tilt images of the blue pick
2 Gaussian map
seed: K = 1,056
(with the C6 copies)
closed-form transform:
F(q) = ∑k ak
× exp(−2π² q⊤ Σk q)
× exp(−2πi q · μk)
3 Pose search
observed −3°
predicted
r = 0.17 at 16 Å
slice × (CTF · dose)²
4 Resolution ladder
12 Å
K 2,418
13.69 Å
10 Å
K 4,296
9.62 Å
split
5.5 Å
K 25,998
5.67 Å
5 Map fit
Gaussians
map
K = 48,000
3.90 / 4.06 Å
6 Polishing ×4
Δ defocus per tilt (Å)
cycle 1 2 3 4
300
0
−300
−20°
0°
20°
particle motion ×100
3.37, 3.32, 3.31, 3.27 Å
7 Result
STAGS
3.27 / 3.25 Å
RELION-5
3.28 Å
Wall clock from the picks
FSC 0.143 crossing at each stage end (Å, half 1)
14.1
13.7
9.6
6.0
5.7
3.90
3.37
3.32
3.31
3.27
7 4.27 h
1–4 seed + ladder, 89 min
5 prep + fit, 24 min
6 4 cycles, 35–37 min each
0 h
1 h
2 h
3 h
4 h
RELION-5
24.1 h
own logs,
other hardware